

Physics Tools

1.0.3

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Chapter 1

Namespace Index

1.1 Package List

Here are the packages with brief descriptions (if available):

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PhysicsTools.Utility	10

Chapter 2

Hierarchical Index

2.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

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PhysicsTools.Collisions.CollisionIgnorer.Entry	19
MonoBehaviour	
PhysicsTools.Collisions.CollisionTracker	17
PhysicsTools.Rigidbody.ForceFollowTransform	19
PhysicsTools.Rigidbody.RigidbodyCOMOverride	25
PhysicsTools.Transformations.TransformVelocityTracker	26
PhysicsTools.Transformations.TransformVelocityTracker.RotationVelocityTimePair	26
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Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

PhysicsTools.Collisions.CollisionIgnorer	
A simple class that allows provides the necessary methods and helpers to temporarily ignore collisions between a set of Colliders and a Collider.	11
PhysicsTools.Collisions.CollisionTracker	
A simple component that keeps track of collisions using OnCollisionEnter and OnCollisionExit.	17
PhysicsTools.Collisions.CollisionIgnorer.Entry	
An entry for a temporarily ignore collision rule.	19
PhysicsTools.Rigidbody.ForceFollowTransform	
A component that uses forces to make a Rigidbody follow a Transform.	19
PhysicsTools.Rigidbody.RigidbodyCOMOverride	
A component that can be attached to a Rigidbody that allows a specified Transform to override the Rigidbody's center of mass.	25
PhysicsTools.Transformations.TransformVelocityTracker.RotationVelocityTimePair	26
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A simple component that tracks velocity and angular velocity for a Transform that it is attached to.	26
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Chapter 4

Namespace Documentation

4.1 PhysicsTools Namespace Reference

4.2 PhysicsTools.Collisions Namespace Reference

Classes

- class [CollisionIgnorer](#)
A simple class that allows provides the neccesary methods and helpers to temporarily ignore collisions between a set of Colliders and a Collider.
- class [CollisionTracker](#)
A simple component that keeps track of collisions using OnCollisionEnter and OnCollisionExit.

4.3 PhysicsTools.Delegates Namespace Reference

Functions

- delegate void [ActionRef< T >](#) (ref T pItem)
A simple delegate for events where an argument is passed by reference.
- delegate void [ActionRef< VALUE_ONE, T >](#) (VALUE_ONE pValueOne, ref T pItem)
A simple delegate for events where an argument is passed by reference.
- delegate void [ActionRef< VALUE_ONE, VALUE_TWO, T >](#) (VALUE_ONE pValueOne, VALUE_TWO pValueTwo, ref T pItem)
A simple delegate for events where an argument is passed by reference.
- delegate void [ActionRef< VALUE_ONE, VALUE_TWO, VALUE_THREE, T >](#) (VALUE_ONE pValueOne, VALUE_TWO pValueTwo, VALUE_THREE pValueThree, ref T pItem)
A simple delegate for events where an argument is passed by reference.

4.3.1 Function Documentation

4.3.1.1 ActionRef< T >()

```
delegate void PhysicsTools.Delegates.ActionRef< T > (  
    ref T pItem )
```

A simple delegate for events where an argument is passed by reference.

Template Parameters

<i>T</i>	The type of the passed reference.
----------	-----------------------------------

Parameters

<i>pItem</i>	The reference that was passed.
--------------	--------------------------------

4.3.1.2 ActionRef< VALUE_ONE, T >()

```
delegate void PhysicsTools.Delegates.ActionRef< VALUE_ONE, T > (
    VALUE_ONE pValueOne,
    ref T pItem )
```

A simple delegate for events where an argument is passed by reference.

Template Parameters

<i>VALUE_ONE</i>	The type of the first passed value.
<i>T</i>	The type of the passed reference.

Parameters

<i>pValueOne</i>	The first value that was passed.
<i>pItem</i>	The reference that was passed.

4.3.1.3 ActionRef< VALUE_ONE, VALUE_TWO, T >()

```
delegate void PhysicsTools.Delegates.ActionRef< VALUE_ONE, VALUE_TWO, T > (
    VALUE_ONE pValueOne,
    VALUE_TWO pValueTwo,
    ref T pItem )
```

A simple delegate for events where an argument is passed by reference.

Template Parameters

<i>VALUE_ONE</i>	The type of the first passed value.
<i>VALUE_TWO</i>	The type of the second passed value.
<i>T</i>	The type of the passed reference.

Parameters

<i>pValueOne</i>	The first value that was passed.
<i>pValueTwo</i>	The second value that was passed.
<i>pItem</i>	The reference that was passed.

4.3.1.4 ActionRef< VALUE_ONE, VALUE_TWO, VALUE_THREE, T >()

```
delegate void PhysicsTools.Delegates.ActionRef< VALUE_ONE, VALUE_TWO, VALUE_THREE, T > (
    VALUE_ONE pValueOne,
    VALUE_TWO pValueTwo,
    VALUE_THREE pValueThree,
    ref T pItem )
```

A simple delegate for events where an argument is passed by reference.

Template Parameters

<i>VALUE_ONE</i>	The type of the first passed value.
<i>VALUE_TWO</i>	The type of the second passed value.
<i>VALUE_THREE</i>	The type of the third passed value.
<i>T</i>	The type of the passed reference.

Parameters

<i>pValueOne</i>	The first value that was passed.
<i>pValueTwo</i>	The second value that was passed.
<i>pValueThree</i>	The third value that was passed.
<i>pItem</i>	The reference that was passed.

4.4 PhysicsTools.Rigidbody Namespace Reference

Classes

- class [ForceFollowTransform](#)
A component that uses forces to make a Rigidbody follow a Transform.
- class [RigidbodyCOMOverride](#)
A component that can be attached to a Rigidbody that allows a specified Transform to override the Rigidbody's center of mass.

4.5 PhysicsTools.Transformations Namespace Reference

Classes

- class [TransformVelocityTracker](#)
A simple component that tracks velocity and angular velocity for a Transform that it is attached to.

4.6 PhysicsTools.Utility Namespace Reference

Classes

- class **RigidbodyUtility**

A public static class that provides useful Rigidbody utilities.

Chapter 5

Class Documentation

5.1 PhysicsTools.Collisions.CollisionIgnorer Class Reference

A simple class that allows provides the necessary methods and helpers to temporarily ignore collisions between a set of Colliders and a Collider.

Classes

- class [Entry](#)

An entry for a temporarily ignore collision rule.

Public Member Functions

- [CollisionIgnorer](#) (List< Collider > pColliders)
Creates a [CollisionIgnorer](#) that provides easy methods for temporarily ignoring collisions between a collider and pColliders.
- void **Update** ()
Updates the collision ignorer allowing it to timeout any temporarily ignored collisions.
- void [IgnoreCollider](#) (Collider pCollider)
Ignore collisions between all colliders associated with this component and pCollider indefinitely.
- void [IgnoreColliders](#) (IEnumerable< Collider > pColliders)
Ignore collisions between all colliders associated with this component and all colliders in the pColliders enumerable indefinitely.
- void [IgnoreColliderFor](#) (Collider pCollider, float pSeconds)
Ignores collisions between all colliders associated with this component and pCollider for pSeconds.
- void [UnignoreColliderIn](#) (Collider pCollider, float pSeconds)
Unignores collisions between all colliders associated with this component and pCollider in pSeconds.
- void [UnignoreCollider](#) (Collider pCollider)
Unignores collisions between all colliders associated with this component and pCollider.
- void [UnignoreColliders](#) (IEnumerable< Collider > pColliders)
Unignores collisions between all colliders associated with this component and all pColliders in the enumerable.
- void [UnignoreEntry](#) ([Entry](#) pEntry)
Undoes the collision ignore entry.
- void [UnignoreIndex](#) (int pIndex)
Undoes the ignore collision registered in entry index pIndex.

- void **UnignoreAll** ()
Undo all temporarily ignored collisions.
- void **IgnoreRigidbody** (Rigidbody pRigidbody)
Ignores collisions between all colliders associated with this component and the Rigidbody, pRigidbody, indefinitely.
- void **IgnoreRigidbodyFor** (Rigidbody pRigidbody, float pSeconds)
Ignores collisions between all colliders associated with this component and the Rigidbody, pRigidbody, for pSeconds.
- void **UnignoreRigidbodyIn** (Rigidbody pRigidbody, float pSeconds)
Unignores collisions between all colliders associated with this component and the Rigidbody, pRigidbody, in pSeconds.
- void **UnignoreRigidbody** (Rigidbody pRigidbody)
Unignores collisions between all colliders associated with this component and the Rigidbody, pRigidbody.

5.1.1 Detailed Description

A simple class that allows provides the necessary methods and helpers to temporarily ignore collisions between a set of Colliders and a Collider.

Author: Mathew Aloisio

5.1.2 Constructor & Destructor Documentation

5.1.2.1 CollisionIgnorer()

```
PhysicsTools.Collisions.CollisionIgnorer.CollisionIgnorer (
    List< Collider > pColliders )
```

Creates a **CollisionIgnorer** that provides easy methods for temporarily ignoring collisions between a collider and pColliders.

Parameters

<i>pColliders</i>	A List of colliders whose collision with other colliders is being controlled by this class instance.
-------------------	--

5.1.3 Member Function Documentation

5.1.3.1 IgnoreCollider()

```
void PhysicsTools.Collisions.CollisionIgnorer.IgnoreCollider (
    Collider pCollider )
```

Ignore collisions between all colliders associated with this component and pCollider indefinitely.

Parameters

<i>pCollider</i>	
------------------	--

5.1.3.2 IgnoreColliderFor()

```
void PhysicsTools.Collisions.CollisionIgnorer.IgnoreColliderFor (
    Collider pCollider,
    float pSeconds )
```

Ignores collisions between all colliders associated with this component and pCollider for pSeconds.

Parameters

<i>pCollider</i>	
<i>pSeconds</i>	

5.1.3.3 IgnoreColliders()

```
void PhysicsTools.Collisions.CollisionIgnorer.IgnoreColliders (
    IEnumerable< Collider > pColliders )
```

Ignore collisions between all colliders associated with this component and all colliders in the pColliders enumerable indefinitely.

Parameters

<i>pColliders</i>	
-------------------	--

5.1.3.4 IgnoreRigidbody()

```
void PhysicsTools.Collisions.CollisionIgnorer.IgnoreRigidbody (
    Rigidbody pRigidbody )
```

Ignores collisions between all colliders associated with this component and the Rigidbody, pRigidbody, indefinitely.

Parameters

<i>pRigidbody</i>	
-------------------	--

5.1.3.5 IgnoreRigidbodyFor()

```
void PhysicsTools.Collisions.CollisionIgnorer.IgnoreRigidbodyFor (
    Rigidbody pRigidbody,
    float pSeconds )
```

Ignores collisions between all colliders associated with this component and the Rigidbody, *pRigidbody*, for *pSeconds*.

Parameters

<i>pRigidbody</i>	
<i>pSeconds</i>	The number of seconds to ignore collisions for.

5.1.3.6 UnignoreCollider()

```
void PhysicsTools.Collisions.CollisionIgnorer.UnignoreCollider (
    Collider pCollider )
```

Unignores collisions between all colliders associated with this component and *pCollider*.

Parameters

<i>pCollider</i>	
------------------	--

5.1.3.7 UnignoreColliderIn()

```
void PhysicsTools.Collisions.CollisionIgnorer.UnignoreColliderIn (
    Collider pCollider,
    float pSeconds )
```

Unignores collisions between all colliders associated with this component and *pCollider* in *pSeconds*.

Parameters

<i>pCollider</i>	
<i>pSeconds</i>	

5.1.3.8 UnignoreColliders()

```
void PhysicsTools.Collisions.CollisionIgnorer.UnignoreColliders (
    IEnumerable< Collider > pColliders )
```

Unignores collisions between all colliders associated with this component and all pColliders in the enumerable.

Parameters

<i>pColliders</i>	
-------------------	--

5.1.3.9 UnignoreEntry()

```
void PhysicsTools.Collisions.CollisionIgnorer.UnignoreEntry (
    Entry pEntry )
```

Undoes the collision ignore entry.

Parameters

<i>pEntry</i>	
---------------	--

5.1.3.10 UnignoreIndex()

```
void PhysicsTools.Collisions.CollisionIgnorer.UnignoreIndex (
    int pIndex )
```

Undoes the ignore collision registered in entry index pIndex.

Parameters

<i>pIndex</i>	
---------------	--

5.1.3.11 UnignoreRigidbody()

```
void PhysicsTools.Collisions.CollisionIgnorer.UnignoreRigidbody (
    Rigidbody pRigidbody )
```

Unggnores collisions between all colliders associated with this component and the Rigidbody, pRigidbody.

Parameters

<i>pRigidbody</i>	
-------------------	--

5.1.3.12 UnignoreRigidbodyIn()

```
void PhysicsTools.Collisions.CollisionIgnorer.UnignoreRigidbodyIn (
    Rigidbody pRigidbody,
    float pSeconds )
```

Unignores collisions between all colliders associated with this component and the Rigidbody, pRigidbody, in pSeconds.

Parameters

<i>pRigidbody</i>	
<i>pSeconds</i>	The number of seconds to unignore collisions in.

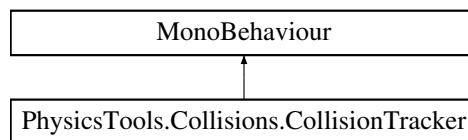
The documentation for this class was generated from the following file:

- CollisionIgnorer.cs

5.2 PhysicsTools.Collisions.CollisionTracker Class Reference

A simple component that keeps track of collisions using OnCollisionEnter and OnCollisionExit.

Inheritance diagram for PhysicsTools.Collisions.CollisionTracker:



Public Member Functions

- bool [IsCollidingWith](#) (Collider pCollider)
Returns true when the tracker is tracking a collision with the specified collider, pCollider, otherwise false.
- bool [IsCollidingWith](#) (Rigidbody pRigidbody, bool pAllowNestedRigidbody)
Returns true if the tracker is colliding with any colliders of pRigidbody, otherwise false.
- Collider [GetCollidingWith](#) (int pIndex)
Returns the Collider in the given 'colliding with' index.

Properties

- int **CollidingWithCount** [get]
Returns the number of Colliders currently colliding with the relevant Rigidbody.

5.2.1 Detailed Description

A simple component that keeps track of collisions using OnCollisionEnter and OnCollisionExit.

Author: Mathew Aloisio

5.2.2 Member Function Documentation

5.2.2.1 GetCollidingWith()

```
Collider PhysicsTools.Collisions.CollisionTracker.GetCollidingWith (
    int pIndex )
```

Returns the Collider in the given 'colliding with' index.

Parameters

<i>pIndex</i>	
---------------	--

Returns

The collider in the given 'colliding with' index.

5.2.2.2 IsCollidingWith() [1/2]

```
bool PhysicsTools.Collisions.CollisionTracker.IsCollidingWith (
    Collider pCollider )
```

Returns true when the tracker is tracking a collision with the specified collider, *pCollider*, otherwise false.

Parameters

<i>pCollider</i>	
------------------	--

Returns

5.2.2.3 IsCollidingWith() [2/2]

```
bool PhysicsTools.Collisions.CollisionTracker.IsCollidingWith (
    Rigidbody pRigidbody,
    bool pAllowNestedRigidbodies )
```

Returns true if the tracker is colliding with any colliders of *pRigidbody*, otherwise false.

Parameters

<i>pRigidbody</i>	
<i>pAllowNestedRigidbody</i>	Should nested Rigidbody colliders be included?

Returns

true if the tracker is colliding with any colliders of pRigidbody, otherwise false.

The documentation for this class was generated from the following file:

- CollisionTracker.cs

5.3 PhysicsTools.Collisions.CollisionIgnorer.Entry Class Reference

An entry for a temporarily ignore collision rule.

Public Attributes

- Collider **collider**
The Collider associated with the entry.
- float **expireTime**
The time the entry is set to expire.

5.3.1 Detailed Description

An entry for a temporarily ignore collision rule.

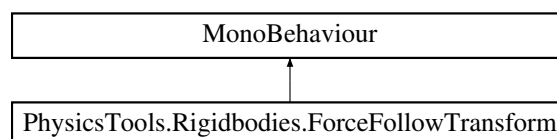
The documentation for this class was generated from the following file:

- CollisionIgnorer.cs

5.4 PhysicsTools.Rigidbody.ForceFollowTransform Class Reference

A component that uses forces to make a Rigidbody follow a Transform.

Inheritance diagram for PhysicsTools.Rigidbody.ForceFollowTransform:



Public Types

- enum **MaxDistanceExceedMode**

Public Member Functions

- void **SetFollowTransform** (Transform pTransform)
A public method that allows the 'followTransform' field of this component to be set. Useful for use in Unity editor events.
- void **SetMaxFollowStrength** (float pStrength)
A public method that allows the 'maxFollowStrength' field of this component to be set. Useful for use in Unity editor events.
- void **SetMaxFollowTorque** (float pTorque)
A public method that allows the 'maxFollowTorque' field of this component to be set. Useful for use in Unity editor events.
- void **SetMaxAngularVelocity** (float pMagnitude)
A public method that allows the 'maxAngularVelocity' field of this component to be set. Useful for use in Unity editor events.
- void **SetMaxFollowDistance** (float pDistance)
A public method that allows the 'maxFollowDistance' field of this component to be set. Useful for use in Unity editor events.
- void **SetMovementEnabled** (bool pEnabled)
A public method that allows the 'enableMovement' field of this component to be set. Useful for use in Unity editor events.
- void **SetRotationEnabled** (bool pEnabled)
A public method that allows the 'enableRotation' field of this component to be set. Useful for use in Unity editor events.
- void **SetPosition** (Vector3 pPosition)
Forcefully sets the position of the Rigidbody.
- void **SetRotation** (Quaternion pRotation)
Forcefully sets the rotation of the Rigidbody.
- void **SetEulerAngles** (Vector3 pEulerAngles)
Invokes SetRotation(Quaternion.Euler(pEulerAngles)).
- void **SetIsKinematic** (bool pKinematic)
A public method that sets the 'isKinematic' field of this component. Useful for use in Unity editor events.

Public Attributes

- bool **enableMovement** = true
Enable movement forces by this component?
- bool **enableRotation** = true
Enable rotation forces by this component?
- bool **isKinematic**
Is this component in kinematic mode?
- Transform **followTransform**
A reference to the Transform this component
- Vector3 **followOffset**
A local space offset to apply to the follow transform position.
- Vector3 **followAngleOffset**
A euler angle offset to apply to the follow transform rotation.
- float **maxFollowStrength** = 750f
The maximum strength to follow the followTransform with.

- float **maxFollowTorque** = 750f
The rotational torque to copy follow rotation with.
- float **maxAngularVelocity** = 10f
The maximum angular velocity for the relevant Rigidbody.
- float **maxFollowDistance** = 2f
The maximum follow distance before the rigidbody performs its
- MaxDistanceExceedMode **distanceExceedMode** = MaxDistanceExceedMode.None
The maximum distance exceeded mode
- PIDController **xMovePID** = new PIDController(5000f, 0f, 500f)
The x
- PIDController **yMovePID** = new PIDController(5000f, 0f, 500f)
The y
- PIDController **zMovePID** = new PIDController(5000f, 0f, 500f)
The z
- PIDRotationController **rotationPID** = new PIDRotationController(2000f, 0f, 30f)
The rotation PID controller.
- UnityEvent **PostMoved**
Invoked every frame after a Move
- UnityEvent **PostRotated**
Invoked every frame after a Rotate
- UnityEvent **MaxFollowDistanceReached**
An event that is invoked when the maximum follow distance is reached and the Rigidbody is snapped back into place.

Properties

- Rigidbody **Rigidbody** [get]
A reference to the Rigidbody associated with this component.
- Vector3 **FollowPosition** [get]
Returns the world space position being targetted.
- Quaternion **FollowRotation** [get]
Returns the world space rotation being targetted.
- float **FollowDistance** [get]
Returns the distance between followTransform.position and transform.position. This always returns the true, unspoofed value (the calculated follow distance) ignoring 'OverrideFollowDistanceDelegate'.
- float **DistanceFactor** [get]
Returns the unclamped distance factor (FollowDistance / maxFollowDistance) for this component.
- float **LastNonKinematicTime** [get]
The last time that this RigidbodyFollowTransform was non-kinematic.

Events

- ActionRef< float > **OverrideFollowDistanceDelegate**
A delegate event that allows the 'FollowDistance' for this component to be spoofed. Arg0: ref float - A reference to the float that is the 'follow distance' that may be spoofed.
- ActionRef< bool > **SkipMoveDelegate**
A delegate event that allows a move to be skipped by an event listener flipping the boolean to true. Arg0: ref bool - Skip the move? If true skip, otherwise don't.
- ActionRef< bool > **SkipRotateDelegate**
A delegate event that allows a rotate to be skipped by an event listener flipping the boolean to true. Arg0: ref bool - Skip the rotate? If true skip, otherwise don't.

5.4.1 Detailed Description

A component that uses forces to make a Rigidbody follow a Transform.

Author: Mathew Aloisio

5.4.2 Member Function Documentation

5.4.2.1 SetEulerAngles()

```
void PhysicsTools.Rigidbodyes.ForceFollowTransform.SetEulerAngles (
    Vector3 pEulerAngles )
```

Invokes SetRotation(Quatenrion.Euler(pEulerAngles)).

Parameters

<i>pEulerAngles</i>	
---------------------	--

5.4.2.2 SetFollowTransform()

```
void PhysicsTools.Rigidbodyes.ForceFollowTransform.SetFollowTransform (
    Transform pTransform )
```

A public method that allows the 'followTransform' field of this component to be set. Useful for use in Unity editor events.

Parameters

<i>pTransform</i>	
-------------------	--

5.4.2.3 SetIsKinematic()

```
void PhysicsTools.Rigidbodyes.ForceFollowTransform.SetIsKinematic (
    bool pKinematic )
```

A public method that sets the 'isKinematic' field of this component. Useful for use in Unity editor events.

Parameters

<i>pKinematic</i>	
-------------------	--

5.4.2.4 SetMaxAngularVelocity()

```
void PhysicsTools.Rigidbody.ForceFollowTransform.SetMaxAngularVelocity (
    float pMagnitude )
```

A public method that allows the 'maxAngularVelocity' field of this component to be set. Useful for use in Unity editor events.

Parameters

<i>pMagnitude</i>	
-------------------	--

5.4.2.5 SetMaxFollowDistance()

```
void PhysicsTools.Rigidbody.ForceFollowTransform.SetMaxFollowDistance (
    float pDistance )
```

A public method that allows the 'maxFollowDistance' field of this component to be set. Useful for use in Unity editor events.

Parameters

<i>pDistance</i>	
------------------	--

5.4.2.6 SetMaxFollowStrength()

```
void PhysicsTools.Rigidbody.ForceFollowTransform.SetMaxFollowStrength (
    float pStrength )
```

A public method that allows the 'maxFollowStrength' field of this component to be set. Useful for use in Unity editor events.

Parameters

<i>pStrength</i>	
------------------	--

5.4.2.7 SetMaxFollowTorque()

```
void PhysicsTools.Rigidbody.ForceFollowTransform.SetMaxFollowTorque (
    float pTorque )
```

A public method that allows the 'maxFollowTorque' field of this component to be set. Useful for use in Unity editor events.

Parameters

<i>pTorque</i>	
----------------	--

5.4.2.8 SetMovementEnabled()

```
void PhysicsTools.Rigidbodyes.ForceFollowTransform.SetMovementEnabled (
    bool pEnabled )
```

A public method that allows the 'enableMovement' field of this component to be set. Useful for use in Unity editor events.

Parameters

<i>pEnabled</i>	
-----------------	--

5.4.2.9 SetPosition()

```
void PhysicsTools.Rigidbodyes.ForceFollowTransform.SetPosition (
    Vector3 pPosition )
```

Forcefully sets the position of the Rigidbody.

Parameters

<i>pPosition</i>	
------------------	--

5.4.2.10 SetRotation()

```
void PhysicsTools.Rigidbodyes.ForceFollowTransform.SetRotation (
    Quaternion pRotation )
```

Forcefully sets the rotaiton of the Rigidbody.

Parameters

<i>pRotation</i>	
------------------	--

5.4.2.11 SetRotationEnabled()

```
void PhysicsTools.Rigidbody.ForceFollowTransform.SetRotationEnabled (
    bool pEnabled )
```

A public method that allows the 'enableRotation' field of this component to be set. Useful for use in Unity editor events.

Parameters

<i>pEnabled</i>	
-----------------	--

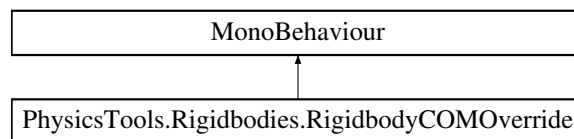
The documentation for this class was generated from the following file:

- ForceFollowTransform.cs

5.5 PhysicsTools.Rigidbody.RigidbodyCOMOverride Class Reference

A component that can be attached to a Rigidbody that allows a specified Transform to override the Rigidbody's center of mass.

Inheritance diagram for PhysicsTools.Rigidbody.RigidbodyCOMOverride:



Public Member Functions

- void **OverrideCenterOfMass** ()
Sets Rigidbody.centerOfMass equal to comTransform.position (in local space of the Rigidbody).
- void **ResetCenterOfMass** ()
resets the related Rigidbody's center of mass to the automatically calculated value.

Public Attributes

- Transform **comTransform**
A reference to the Transform that represents the location of the Rigidbody
- bool **overrideOnAwake** = true
Should the CoM be automatically overridden on

Properties

- Rigidbody **Rigidbody** [get]
A reference to the Rigidbody associated with this component.

5.5.1 Detailed Description

A component that can be attached to a Rigidbody that allows a specified Transform to override the Rigidbody's center of mass.

Author: Intuitive Gaming Solutions

The documentation for this class was generated from the following file:

- RigidbodyCOMOverride.cs

5.6 PhysicsTools.Transformations.TransformVelocityTracker.RotationVelocityTimePair Struct Reference

Public Attributes

- Quaternion **rotationVelocity**
- float **time**

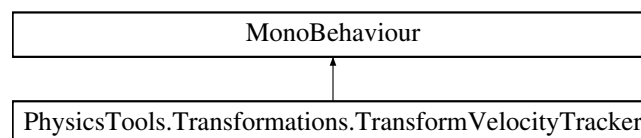
The documentation for this struct was generated from the following file:

- TransformVelocityTracker.cs

5.7 PhysicsTools.Transformations.TransformVelocityTracker Class Reference

A simple component that tracks velocity and angular velocity for a Transform that it is attached to.

Inheritance diagram for PhysicsTools.Transformations.TransformVelocityTracker:



Classes

- struct [RotationVelocityTimePair](#)
- struct [VelocityTimePair](#)

Public Member Functions

- void **ClearData** ()
Clears both velocity and angular velocity data.
- void **ClearVelocityData** ()
Clears all velocity data.
- void **ClearAngularVelocityData** ()
Clears all angular velocity data.
- Vector3 **GetAverageVelocityOverSeconds** (float pSeconds)
Computes the average velocity for the [TransformVelocityTracker](#) over the last pSeconds seconds. Adding more seconds past the length of data history the component tracks has no effect.
- Quaternion **GetAverageRotationVelocityOverSeconds** (float pSeconds)
Computes the average rotation velocity using tracked data from this component over the last pSeconds seconds. If the component does not have that many seconds of data then increasing pSeconds will have no effect.
- void **SetSampleVelocityEnabled** (bool pEnabled)
Sets the 'sampleVelocity' field of this component. Useful for use with Unity editor events.
- void **SetSampleRotationVelocityEnabled** (bool pEnabled)
Sets the 'sampleRotationVelocity' field of this component. Useful for use with Unity editor events.
- void **SetMoveSampleTime** (float pTime)
Sets the 'moveSampleTime' field of this component. Useful for use with Unity editor events.
- void **SetRotateSampleTime** (float pTime)
Sets the 'rotateSampleTime' field of this component. Useful for use with Unity editor events.

Public Attributes

- bool **sampleVelocity** = true
Is velocity sampling enabled for this component?
- float **moveSampleTime** = 0.4f
All velocity samples up to velocitySampleTime seconds old will be used in average velocity calculations.
- bool **sampleRotationVelocity** = true
Is rotation velocity sampling enabled for this component?
- float **rotateSampleTime** = 0.1f
All angular/rotational velocity samples up to angularVelocitySampleTime seconds old will be used in average angular/rotational velocity calculations.

Properties

- Vector3 **PositionLastFrame** [get]
The position of the Grabber's transform last frame.
- Quaternion **RotationLastFrame** [get]
The rotation of the Grabber's transform last frame.
- Vector3 **AverageVelocity** [get]
Returns the average velocity of the Transform in units per second over the last moveSampleTime seconds.
- int **AverageVelocityEntries** [get]
Returns the number of velocity entries used in the last 'AverageVelocity' calculation.
- Vector3 **VelocityLastFrame** [get]
The velocity in units per second last frame.
- Quaternion **AverageRotationVelocity** [get]
Returns the average rotation velocity of the Transform in units per second over the last rotateSampleTime seconds.
- int **AverageRotationVelocityEntries** [get]
Returns the number of rotation velocity entries used in the last 'AverageRotationVelocity' calculation.
- Quaternion **RotationVelocityLastFrame** [get]
The rotational velocity last frame represented as a Quaternion.

5.7.1 Detailed Description

A simple component that tracks velocity and angular velocity for a Transform that it is attached to.

Author: Mathew Aloisio

5.7.2 Member Function Documentation

5.7.2.1 GetAverageRotationVelocityOverSeconds()

```
Quaternion PhysicsTools.Transformations.TransformVelocityTracker.GetAverageRotationVelocity↔  
OverSeconds (   
    float pSeconds )
```

Computes the aerge rotation velocity using tracked data from this component over the last pSeconds seconds. If the component does not have that many seconds of data then increasing pSeconds will have no effect.

Parameters

<i>pSeconds</i>	
-----------------	--

Returns

the aerge rotation velocity using tracked data from this component over the last pSeconds seconds.

5.7.2.2 GetAverageVelocityOverSeconds()

```
Vector3 PhysicsTools.Transformations.TransformVelocityTracker.GetAverageVelocityOverSeconds (   
    float pSeconds )
```

Computes the average velocity for the [TransformVelocityTracker](#) over the last pSeconds seconds. Adding more seconds past the length of data history the component tracks has no effect.

Parameters

<i>pSeconds</i>	
-----------------	--

Returns

the average velocity for the [TransformVelocityTracker](#) over the last pSeconds seconds.

5.7.2.3 SetMoveSampleTime()

```
void PhysicsTools.Transformations.TransformVelocityTracker.SetMoveSampleTime (
    float pTime )
```

Sets the 'moveSampleTime' field of this component. Useful for use with Unity editor events.

Parameters

<i>pTime</i>	
--------------	--

5.7.2.4 SetRotateSampleTime()

```
void PhysicsTools.Transformations.TransformVelocityTracker.SetRotateSampleTime (
    float pTime )
```

Sets the 'rotateSampleTime' field of this component. Useful for use with Unity editor events.

Parameters

<i>pTime</i>	
--------------	--

5.7.2.5 SetSampleRotationVelocityEnabled()

```
void PhysicsTools.Transformations.TransformVelocityTracker.SetSampleRotationVelocityEnabled (
    bool pEnabled )
```

Sets the 'sampleRotationVelocity' field of this component. Useful for use with Unity editor events.

Parameters

<i>pEnabled</i>	
-----------------	--

5.7.2.6 SetSampleVelocityEnabled()

```
void PhysicsTools.Transformations.TransformVelocityTracker.SetSampleVelocityEnabled (
    bool pEnabled )
```

Sets the 'sampleVelocity' field of this component. Useful for use with Unity editor events.

Parameters

<i>pEnabled</i>	
-----------------	--

The documentation for this class was generated from the following file:

- TransformVelocityTracker.cs

5.8 PhysicsTools.Transformations.TransformVelocityTracker.Velocity↔ TimePair Struct Reference

Public Attributes

- Vector3 **velocity**
- float **time**

The documentation for this struct was generated from the following file:

- TransformVelocityTracker.cs

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